

DATA MANAGEMENT PLAN

Multi-Modal Agentic AI System for Medical Diagnosis and Triage

Team GitHub: <https://github.com/abhishek7112000/Multi-Modal-Agentic-AI-System-for-Medical-Diagnosis-and-Triage>

Team Member: Ajayi Israel

Team Role: Communications leader

1. OVERVIEW OF DATASETS

Three datasets are used across the three required ML models, all focused on chest and cardiovascular conditions — among the UK's leading causes of mortality.

Dataset 1 — NIH ChestX-ray14 | Model 1: ChestVision (Custom NN)

- **Format/Size:** 112,120 PNG images (1024×1024 px, greyscale) + Data_Entry_2017.csv metadata. ~45 GB total (12 compressed archives).
- **Structure:** 30,805 de-identified patients; 14 thoracic disease labels (Cardiomegaly, Pneumonia, Effusion, Infiltration, Atelectasis, Edema, etc.); official patient-level train/val/test splits.
- **Suitability:** Directly supplies labelled chest X-ray images for training ChestVision to perform 14-class thoracic disease classification.

Dataset 2 — Synthea Synthetic EHR | Model 2: ClinicalFusion (Hybrid)

- **Format/Size:** 6 CSV files (patients, encounters, conditions, observations, medications, procedures). ~500 MB compressed / ~2 GB uncompressed.
- **Structure:** 1,000+ synthetic patients; relational schema linked by patient ID. Includes ICD-10 diagnoses, LOINC lab codes (troponin, BNP, WBC, SpO₂, CRP), vitals, medications, encounter notes. Filtered to ICD-10: I21, I25, I50, I48, I10, J18, J44, J45, J96.
- **Suitability:** Provides 60+ engineered structured features and encounter note text for training ClinicalFusion ((MLP tabular branch + BioClinical-ModernBERT-large text branch) to predict diagnosis category and severity score (0–3).

Dataset 3 — Malikeh1375/medical-question-answering-datasets | Model 3: OpenMed NER (Transfer Learning)

- **Format/Size:** Parquet/JSON files via HuggingFace datasets library. ~2.3 GB. Aggregates MedQA, MedMCQA, PubMedQA, and HealthSearchQA.
- **Structure:** Fields: question, answer, options (where applicable), context, source. Language: English. Rich in clinical disease names, anatomy, and pharmaceutical terminology.
- **Suitability:** Provides diverse biomedical text for fine-tuning OpenMed-NER-SuperClinical-434M on chest/cardiovascular entity extraction (B-DISEASE, B-ANATOMY, B-PHARMA tags).

Knowledge Base (not a training dataset): A-Z Family Medical Encyclopedia is chunked, embedded using the OpenMed NER encoder, and stored in ChromaDB for RAG QA enrichment post-diagnosis. Not used in model training.

2. DATA COLLECTION PLAN

- **Dataset 1 — NIH ChestX-ray14:** Direct download from <https://nihcc.app.box.com/v/ChestXray-NIHCC> (no registration). 12 .tar.gz archives + CSV files. Provider: NIH Clinical Center, USA. Citation: Wang et al. (2017), IEEE CVPR.
- **Dataset 2 — Synthea EHR:** Download from <https://www.kaggle.com/datasets/imtkaggleteam/synthetic-medical-dataset> (free Kaggle account). Single compressed archive. Citation: Walone et al. (2017), JAMIA 25(3):230–238.
- **Dataset 3 — Medical QA:** Loaded programmatically via HuggingFace: `load_dataset('Malikeh1375/medical-question-answering-datasets')`. No manual download. URL: <https://huggingface.co/datasets/Malikeh1375/medical-question-answering-datasets>. Licence: Apache 2.0.

Download dates, source URLs, checksums, and any preprocessing steps are recorded in /data/metadata/collection_log.md in the GitHub repository.

3. METADATA

- **Dataset 1 — NIH:** Image_Index, Finding_Labels (pipe-separated), Patient_ID, Patient_Age, Patient_Gender (M/F), View_Position (PA/AP), Image_Width, Image_Height. BBox file: Image_Index, Finding_Label, x, y, w, h (used for Grad-CAM localisation).
- **Dataset 2 — Synthea:** Patients: ID, BIRTHDATE, GENDER, RACE, ETHNICITY, CITY, ZIP. Conditions: PATIENT, CODE (ICD-10/SNOMED), DESCRIPTION, START, STOP. Observations: CODE (LOINC), DESCRIPTION, VALUE, UNITS, DATE. Encounters: ID, ENCOUNTERCLASS, REASONDESCRIPTION.

- **Dataset 3 — Medical QA:** question (string), answer (string), options (list), context (string, optional), source (subset name). Apache 2.0. Language: English.

Full metadata dictionaries (field definitions, value ranges, missing-value strategies, imputation approach) are stored in /data/metadata/ in the GitHub repository.

4. DOCUMENT CONTROL

GitHub: <https://github.com/abhishek7112000/Multi-Modal-Agentic-AI-System-for-Medical-Diagnosis-and-Triage>

Repository Structure

- /data — /raw (gitignored), /processed, /metadata, /splits
- /notebooks — /01_eda, /02_preprocessing, /03_training, /04_evaluation
- /src — /data_processing, /models, /agent, /api, /app
- /models — /onnx, /quantised (checkpoints gitignored); /docs; /tests
- Root: README.md · .gitignore · requirements.txt · package.json · .env.example

File Naming Convention

- **Datasets:** {name}_{version}_{YYYYMMDD}.{ext} → nih_chestxray_v1_20250901.csv
- **Scripts/Notebooks:** {nn}_{purpose}_{dataset}.py/.ipynb → 02_preprocess_synthea_ehr.ipynb
- **Models:** {model}_{arch}_{version}_{YYYYMMDD}.onnx → chestvision_resnet50_v2_20251101.onnx

Version Control & README

- Branches: main (production) · develop (integration) · feature/* · model/*. Commit format: [TYPE] description. PRs require one reviewer sign-off before merging to main.
- README.md covers: project overview; team members; dataset sources and access; setup instructions (Python 3.11, Node.js 20); training and export commands; app launch; CI/CD status badge; licence information.

5. SECURITY AND STORAGE

- **Compute & Training:** University of Hertfordshire computing laboratory — high-specification workstations with dedicated GPUs for all model training (Models 1–3), ONNX export, and quantisation.
- **Primary Storage:** UH lab local drives (~100 GB for raw + processed data). Nightly automated backup to UH network storage.
- **Code & Versioning:** GitHub repository (private during development; public post-grading). Large datasets and model checkpoints gitignored.
- **Production:** Vercel (Next.js + ONNX API routes) · Railway (ChromaDB) · NeonDB (serverless PostgreSQL) — all under UH team accounts.
- **Security:** Raw data never committed to Git. API keys (.env) gitignored and shared via UH encrypted drive. Branch protection on main. No real patient data at any stage.
- **Retention:** All data kept September 2025 – May 2026. Post-project: raw datasets 2 years; models archived permanently on UH network. Secure cloud deletion after 1-year retention.

6. ETHICAL STATEMENT

Q1: Does the data come under GDPR requirements?

No. NIH ChestX-ray14 is fully de-identified under HIPAA Safe Harbor — all PII removed (names, dates, contact details, device IDs); ages in years only; patients over 89 grouped. No identifying embedded image metadata. Released by NIH after expert de-identification review. Synthea EHR is entirely computer-generated — no real individuals exist, so GDPR is inapplicable. Medical QA (Malikeh1375) aggregates publicly available NLP corpora (medical licensing exam questions, PubMed abstracts, consumer health queries) — no identifiable patient records. No special categories of personal health data relating to any living individual are processed at any stage.

Q2: Does the project conform to UH ethical policies?

Yes. No primary data collection from human participants is conducted — no surveys, interviews, or direct observations. All data is secondary, from public repositories. Per UH Research Ethics Committee guidelines, projects using only publicly available, fully de-identified or synthetic datasets for educational purposes do not require formal ethical review. The team commits to: using data solely for stated educational research; not attempting re-identification; citing all sources; following UH Responsible AI principles; documenting per-demographic bias monitoring for ChestVision and ClinicalFusion; including a clinical safety disclaimer in all outputs (decision-support tool, not a clinical replacement); implementing doctor human-in-the-loop review and override for all triage decisions. Supervisor oversight maintained throughout.

Q3: Do you have permission to use the data for your proposed research project?

Yes. NIH ChestX-ray14: Permissive public use licence — 'made available to the scientific community for educational and research purposes.' Attribution: Wang et al. (2017). Synthea EHR: Apache Licence 2.0 — unrestricted use for research and education; attribution to MITRE/Synthea. Medical QA (Malikeh1375): Apache Licence 2.0 — explicitly permits NLP and ML research use; confirmed on HuggingFace dataset card. All three licence files are archived in /docs/data_licenses/ in the GitHub repository with screenshots as evidence.

Q4: Are you assured that the data was collected ethically?

Yes. NIH ChestX-ray14: Collected by NIH Clinical Center (a U.S. government research hospital) during routine clinical care (1992–2015). Original collection approved by NIH Institutional Review Board (IRB); retrospective research use authorised under IRB waiver (minimal risk, de-identified). De-identification by NIH experts per HIPAA. Published peer-reviewed (IEEE CVPR 2017); no ethical concerns in 7+ years of widespread academic use. Synthea EHR: No human data collected — Synthea algorithm generates fictional patients from aggregated population statistics and published clinical guidelines (not individual records). Open-source, fully auditable methodology (JAMIA 2017). Developed by MITRE Corporation (federally funded; strong ethical governance); purpose-built to eliminate patient privacy risk. Medical QA: Aggregates public sources — MedQA (medical licensing exams), PubMedQA (open-access literature), HealthSearchQA (anonymised consumer queries). No private records used; full provenance documented on HuggingFace. All evidence (IRB references, licences, dataset cards, citations) archived in /docs/data_licenses/ in the GitHub repository.

Conclusion: No real patient data is used at any stage. All training datasets are de-identified public imaging (NIH) or entirely synthetic/public text (Synthea, Medical QA). No GDPR obligations arise; all usage permissions are held; all datasets originate from ethically reviewed processes. The project fully complies with University of Hertfordshire ethical requirements for student research projects.